UNITED ARAB EMIRATES MINISTRY OF EDUCATION
الإمارات العربية المتحدة وزارة التربية والتعليم

## Al Khalidiya Boys School – Cycle 3

School code 10048 | Dibba Al Hisn | Sharjah Eastern

# Grade 11 Physics Final Exam Practice

**Term 1 2026–2027 • Set C • With Answers**

**100 marks | 150 minutes | Calculator allowed**

Answer all 20 questions. Choose one option for each MCQ. Show working, units and labelled diagrams for Q17–20. Use g = 9.8 m/s² when needed.

## Section A Multiple Choice 60 marks

**1. Earth rotates once in 24 h. Through what angle does it rotate in 6 h? [4 marks]**

A. π/4 rad
B. 2π rad
**C. π/2 rad**
D. π rad

Reference: p. 4 | **Answer C**

**2. A disk rotates at constant ω. Which point has the greatest tangential speed? [4 marks]**

**A. A point on its outer rim**
B. Every point has the same speed
C. Its centre
D. The point nearest its centre

Reference: p. 5 | **Answer A**

**3. For tangential acceleration a at radius r, which relation is correct? [4 marks]**

A. $a = \alpha/r$
**B. $a = r\alpha$**
C. $a = r^2\alpha^2$
D. $a = r/\alpha$

Reference: p. 5 | **Answer B**

**4. Which relationship connects angular speed and frequency? [4 marks]**

A. $\omega = f/(2\pi)$
B. $\omega = f^2$
C. $\omega = 2\pi/f$
**D. $\omega = 2\pi f$**

Reference: pp. 5–6 | **Answer D**

**5. In one revolution, a point twice as far from the axis travels: [4 marks]**

A. Half the distance
B. The same distance
**C. Twice the distance**
D. Four times the distance

Reference: p. 4 | **Answer C**

**6. Taking counterclockwise as positive, equal clockwise and counterclockwise torques give: [4 marks]**

A. Twice either torque
B. Positive net torque
C. Negative net torque
**D. Zero net torque**

Reference: pp. 8–9 | **Answer D**

**7. If a point mass is moved to twice its original distance from an axis, its moment of inertia becomes: [4 marks]**

**A. Four times as large**
B. Twice as large
C. Unchanged
D. Half as large

Reference: p. 12 | **Answer A**

**8. Which design generally makes an object more stable? [2 marks]**

**A. A lower centre of mass and wider base**
B. A higher centre of mass and narrower base
C. A narrower base only
D. A higher centre of mass only

Reference: pp. 18–20; p. 24 | **Answer A**

**9. When choosing a pivot for an equilibrium calculation, which choice is often useful? [4 marks]**

A. Only a point outside the object
B. A point with no physical location
**C. A point where an unknown force acts**
D. Only the centre of Earth

Reference: pp. 20–21 | **Answer C**

**10. In the Northern Hemisphere, the Coriolis deflection of horizontal motion is toward the: [4 marks]**

A. Left of the direction of motion
**B. Right of the direction of motion**
C. Same direction as the motion only
D. Centre of the moving object

Reference: pp. 23–24 | **Answer B**

**11. Which equation follows by combining F = ma with constant-mass momentum? [4 marks]**

**A. $F\,\Delta t = \Delta p$**
B. $F = p\,\Delta t$
C. $\Delta p = m/\Delta t$
D. $F\,\Delta p = \Delta t$

Reference: pp. 30–31 | **Answer A**

**12. A 7.0 kg object receives +5.0 N for 1.0 s followed by −5.0 N for 1.0 s. Its total momentum change is: [4 marks]**

A. +5 kg m/s
B. +10 kg m/s
C. −10 kg m/s
**D. 0 kg m/s**

Reference: p. 33, Practice Problem 3 | **Answer D**

**13. An astronaut fires gas backward in space. With negligible external impulse, the astronaut moves forward because: [4 marks]**

A. The gas pushes on surrounding air
**B. Total momentum of astronaut and gas is conserved**
C. Momentum is destroyed in the gas
D. The astronaut gains mass

Reference: p. 42, Example Problem 4 | **Answer B**

**14. Why do padded surfaces reduce injury for the same initial and final momentum? [4 marks]**

A. They guarantee zero impulse
**B. They spread the momentum change over more time**
C. They stop all interaction forces
D. They increase the rate of momentum change

Reference: p. 33; p. 37, Check Your Progress 11 | **Answer B**

**15. A skater extends the arms while spinning. With negligible external torque, angular speed: [4 marks]**

A. Must reverse direction
B. Increases
**C. Decreases**
D. Becomes infinite

Reference: pp. 34–35 | **Answer C**

**16. A spinning gyroscope resists a change in the direction of its axis because: [2 marks]**

A. Gravity never acts on it
B. Its linear momentum must be zero
C. It has no moment of inertia
**D. Changing angular momentum requires external torque**

Reference: pp. 45–47 | **Answer D**

## Set C Section B Free Response

**17. A washer drum rotates at 635 rev/min and stops uniformly in 8.0 s. Separately, a 0.234 m wrench must supply 32.4 N m; the applied force is 232 N. [10 marks]**

Reference: p. 6, Check Your Progress 8; p. 9, Practice Problem 13; pp. 20–21

(a) Convert the initial angular velocity to rad/s. [2]

**Answer** $\omega_i = 635(2\pi/60)$ [1] = 66.5 rad/s [1].

(b) Calculate the drum's angular acceleration, taking its initial rotation as positive. [3]

**Answer** $\alpha = (\omega_f - \omega_i)/t$ [1]; $\alpha = (0 - 66.5)/8$ [1] = −8.31 rad/s² [1].

(c) Find the smallest angle between the applied force and wrench handle that produces the required torque. [3]

**Answer** $\tau = rF\sin\theta$ [1]; $\sin\theta = 32.4/(0.234 \times 232)$ [1]; $\theta = 36.6°$ [1].

(d) For a lever in static equilibrium, state the relationship between clockwise and counterclockwise torques. [2]

**Answer** Their total magnitudes are equal [1]; their signed sum is zero [1].

**18. Three uniform objects each have mass 1.0 kg and radius 2.0 m: a thin hoop, solid cylinder and solid sphere. Use $I_{hoop} = MR^2$, $I_{cylinder} = \frac{1}{2}MR^2$, $I_{sphere} = \frac{2}{5}MR^2$. Separately, a skater with I = 0.50 kg m² spins at 2.5 rev/s and stops in 3.0 s. [10 marks]**

Reference: p. 15, Practice Problem 23; p. 36, Example Problem 2

(a) Calculate I for each of the three objects. [3]

**Answer** Hoop 4.0 kg m² [1]; cylinder 2.0 kg m² [1]; sphere 1.6 kg m² [1].

(b) Under equal net torque, which object has the greatest angular acceleration? Explain. [2]

**Answer** Solid sphere [1]; $\alpha = \tau/I$ and it has the smallest I [1].

(c) Find the skater's initial angular momentum. [2]

**Answer** $\omega = 2.5(2\pi)$ rad/s [1]; $L = I\omega = 7.85$ kg m²/s [1].

(d) Taking the initial spin as positive, calculate angular impulse and average braking torque. [3]

**Answer** $\Delta L = 0 - 7.85 = -7.85$ N m s [1]; $\tau_{average} = \Delta L/\Delta t$ [1] = −2.62 N m [1].

**19. A 0.174 kg softball approaches a bat at +26.0 m/s and leaves at −38.0 m/s. Contact lasts 0.80 ms. [10 marks]**

Reference: p. 37, Check Your Progress 12; pp. 30–32

(a) Calculate the ball's initial and final momenta. [2]

**Answer** $p_i = 0.174(26) = +4.524$ kg m/s [1]; $p_f = 0.174(-38) = -6.612$ kg m/s [1].

(b) Calculate the impulse on the ball, including direction. [3]

**Answer** $J = \Delta p = m(v_f - v_i)$ [1] = −11.136 N s [1], opposite the initial motion [1].

(c) Find the average force on the ball. [3]

**Answer** $\Delta t = 8.0 \times 10^{-4}$ s [1]; $F_{average} = J/\Delta t$ [1] = $-1.39 \times 10^4$ N [1].

(d) State the impulse the ball exerts on the bat and explain. [2]

**Answer** +11.136 N s [1]; equal-magnitude opposite interaction forces act for the same time [1].

**20. A 0.50 kg ball moves at +6.0 m/s toward a 1.00 kg ball moving at −12.0 m/s. After a head-on collision, the 0.50 kg ball has velocity −14 m/s. Neglect external impulse. [10 marks]**

Reference: p. 40, Practice Problem 20; pp. 30–31

(a) Calculate total initial momentum. [2]

**Answer** $p_i = 0.50(6.0) + 1.00(-12.0)$ [1] = −9.0 kg m/s [1].

(b) Find the final velocity of the 1.00 kg ball. [3]

**Answer** $-9 = 0.50(-14) + 1.00v$ [1]; $v = -2.0$ m/s [1], in its original negative direction [1].

(c) Calculate the impulse on each ball. [3]

**Answer** $J_1 = 0.50(-14 - 6) = -10$ N s [1]; $J_2 = 1.00(-2 - (-12)) = +10$ N s [1]; equal and opposite [1].

(d) Explain why each ball's momentum may change while total momentum stays constant. [2]

**Answer** The interaction forces are internal to the two-ball system [1]; their opposite impulses cancel in the total [1].